Names:_____

_____

Section: _____ Bench #:_____

Experiment 8: Linear Power Supply with Audio Amplifier

8.3. Procedure: Mains Voltage, Step-Down Transformer, and Power Module Setup

Measurements: Power Module Output

FREQUENCY: _____ Hz

PEAK-TO-PEAK: _____ V

AMPLITUDE: _____ V

RMS: _____ V

Screen capture reminder: _____

Transformer rated voltage: _____ VAC

Transformer rated current: _____ A

Calculated internal resistance: _____ Ω

Instructor Verification:

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8.4. Procedure: Full-Wave Bridge Rectifier

Measurements: Rectifier Output

Rectifier output frequency: _____ Hz

Rectifier output amplitude: _____ V

Amplitude difference ($V_{OUT} - V_{IN}$): _____ V

Screen capture reminder: _____

Why is the frequency at the output of the rectifier *twice* that of the input from the power module?

The change (drop) in amplitude from input to output indicates that this is not an ideal rectifier. What is causing the extra voltage drop?

8.4.1. Rectifier Filter Capacitors

Measurements: Filter Capacitors

Table 8-1: Rectifier ripple data.

FILTER CAPACITOR:	10 µF	100 µF
MEAN VOLTAGE:		
PK-PK VOLTAGE (RIPPLE):		

Screen capture reminder: _____

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Suppose we decrease the value of R_1 to 330 Ω. What effect would this have on the MEAN value and PEAK TO PEAK ripple at the output? Feel free to test your hypothesis.

Instructor Verification:

8.5. Procedure: Audio Amplifier with Unregulated Supply

Describe the tone of the audio when using the 100 µF capacitor in your unregulated supply.

Describe the overall tone and quality of the audio using the 10 µF capacitor compared to the 100 µF one. **When finished, remove the 10 µF capacitor and put the 100 µF capacitor back in the circuit**.

Instructor Verification:

8.6. Procedure: Active Voltage Regulator

Regulated supply ripple (RMS): _____ V

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Describe the quality of the audio as compared to using the unregulated supply earlier.

Instructor Verification:

8.7. Lab Report

Turn in all pages of recorded data, screenshots, plots, and tables collected during the lab. Be sure to answer all questions throughout the experiment. Clean up the lab bench, remove all circuit parts and wires from the breadboard. Place used components in the used component bins. Return all cables to their proper place. Turn off all lab bench equipment and answer the following questions.

Based on your observations, why would a high-fidelity audio amplifier require an internal high-quality DC power supply?

For a sine wave, how are the RMS and amplitude voltages related? Give the equation. Using this equation, calculate the RMS voltage of the transformer output from the amplitude measured in Section 8.3. Compare your calculated value to the one obtained from the scope's automated measurement.

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